

Delphes selection vs. the CROWN ntuple baseline

the stage the NSBI test's CMS ntuples correspond to

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Abstract

Comparison of the `--crown` selection applied to the Delphes samples against the ntuple-level selection in KIT-CMS/BBTauTauAnalysis-CROWN (`nmssm_config.py`). This is a *different stage* from the published HIG-25-008 analysis selection: CROWN keeps $p_T > 20$ on all three legs and τ_h candidates at the **VVVLoose** working point, applying the trigger thresholds, the elliptical signal region and the categorisation only later. Comparing our analysis-level selection against their ntuple-level one is what produced an apparent factor ~ 5 discrepancy in signal efficiency; the correct comparison is this one.

1 Object selection

	CROWN ntuple	source		Delphes <code>--crown</code>	
Muon $p_T, \eta $	<code>tight_muon_min_pt</code> <code>max_abs_eta 2.4</code>	20,	config	20, 2.4	✓
Electron $p_T, \eta $	<code>tight_electron_min_pt</code> <code>max_abs_eta 2.5</code>	20,	config	20, 2.5	✓
τ_h $p_T, \eta $	<code>tight_tau_min_pt</code> <code>max_abs_eta 2.5</code>	20,	config	20, 2.5	✓
τ_h vs-jet WP	VVVLoose (<code>wp: 1</code>) — “ <i>looser taus needed for tau misidentification estimate</i> ”		config	one Medium -equivalent bit	✗
τ_h vs- e / vs- μ WP	VVVLoose / VLoose (<code>wp: 1</code>)		config	— (single bit)	✗
τ_h decay modes	0, 1, 10, 11 (excludes DM 2)		config	none — Delphes has no decay modes	✗
$\tau_h d_z $	0.2 cm		config	not available	✗
Lepton $ d_{xy} , d_z $	0.045 / 0.2 cm		paper Tab. 1	not available	✗
Lepton ID / isolation	Tight ID + isolation		paper Tab. 1	card efficiency only	✗
b-jet $p_T, \eta $	<code>bjet_min_pt</code> <code>bjet_max_abs_eta 2.5</code> (Run 3)	20,	config	20, 2.5	✓
b-tag requirement	none at ntuple level		inferred [†]	none (<code>btag_min 0</code>)	✓

[†] No b-tag requirement was found in the configuration; `bjet_min_pt` defines the collection rather than imposing a cut.

2 Pair and event selection

3 Where the residual efficiency gap lives

τ_h **working point — the dominant term.** CROWN keeps τ_h candidates at VVVLoose and applies Medium later in the analysis, precisely so the misidentification background can be estimated from the looser sample. A Delphes τ_h carries a *single bit* already fixed at a Medium-equivalent working point: there is no looser candidate to keep, because it was never written. DEEPTAU Medium is $\sim 70\%$ efficient where VVVLoose is $\sim 98\%$, so a factor ~ 1.4 on the τ_h leg is expected and cannot be recovered by any selection change. This is the same root cause as the τ_h -object rebuild on the tuning-v2 critical path.

	CROWN ntuple	source	Delphes --crown	
Channel assignment	$\mu \rightarrow \text{mt}$, else $e \rightarrow \text{et}$, else 2 $\tau_h \rightarrow \text{tt}$	paper §5	same priority	✓
Opposite charge	required	—	not applied: no Jet.charge in these ntuples	✗
$\Delta R(\ell, \tau_h)$	> 0.5	CROWN author	> 0.5 applied	✓
Extra-lepton veto	loose e/μ with $p_T > 10$ (<code>loose_{muon,electron}_min_pt</code>)	config, confirmed	loose e/μ with $p_T > 10$	✓
Elliptical signal region	not applied	—	not applied	✓
Trigger	not at ntuple level	inferred	none	✓

Opposite charge — works the other way. We keep same-sign pairs that CROWN rejects. Same-sign is CMS’s QCD control region at $\sim 99\%$ purity, i.e. an essentially pure fake- τ population, so *our background is inflated* by its absence. Delphes does write `Jet.Charge`; it was simply not carried into the ntuple schema. Now added, and it arrives with the next ntuplisation — no re-simulation required.

Decay modes and impact parameters. Not representable: Delphes has no HPS decay modes and no track impact parameters. Both are acceptance-reducing on the CROWN side, so their absence also inflates our yield slightly.

4 Resolved since the first version of this note

Both open rows were confirmed by the CROWN author and are now matched:

1. $\Delta R(\ell, \tau_h) > 0.5$ **is applied in CROWN**, as we already did — no change needed.
2. **The extra-lepton veto uses loose e/μ at $p_T > 10$** , not the trigger-motivated selection thresholds. Our veto previously counted only leptons above the selection threshold (20), so a second lepton at, say, 15 GeV was invisible to it and the event was kept where CROWN rejects it. Now fixed: the veto runs over loose collections at $p_T > 10$, $|\eta| < 2.4$ (μ) / 2.5 (e). This *tightens* the Delphes selection, so yields quoted before this change are slightly high.

Implementation: `scripts/delphes_to_sbi.py` — `CROWN_SEL` and `cms_select(thresholds=CROWN_SEL)`, reached with `--crown` on both `delphes_to_sbi.py` and `cutflow.py`. Thresholds pinned against the CROWN configuration by tests in `tests/test_cutflow.py`. A companion note, `selection_comparison.tex`, compares the same converter against the published HIG-25-008 *analysis* selection.